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SYSTEM PACKAGING FOR CELLULAR MODEM AND TRANSCEIVER SYSTEM OF HETEROGENEOUS STACKING

Abstract

A radio frequency package includes a baseband processor, a transceiver, and a memory. The baseband processor performs processing for wireless communication functions. Moreover, the transceiver transmits and receives wireless signals based on the processing of the wireless communication functions. Additionally, the memory is associated with the baseband processor and stores instructions for performing the processing. The memory and the baseband processor are disposed on top of the transceiver.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/582,870, filed Jan. 24, 2022, entitled “SYSTEM PACKAGING FOR CELLULAR MODEM AND TRANSCEIVER SYSTEM OF HETEROGENEOUS STACKING,” which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

[0002] The present disclosure relates generally to wireless communication systems and devices and, more specifically, to system packaging that accommodates wireless communication components while conserving space.

[0003] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

[0004] Generally, radio frequency devices may include a package, such as a system-in-package (SiP). A system-in-package incorporates substrates, dies, multiple integrated circuits, and/or passive devices into a single package. The system-in-package may have many components formed using semiconducting material, such as silicon. For example, the substrates and dies may include silicon on which the integrated circuits are fabricated. The substrates, dies, and devices may be coupled by wires bonded to the package or by solder joints (e.g., solder balls or pads). By way of example, the dies may be placed adjacent to each other on a single chip or substrate, as well be stacked on top of one another in the system-in-package.

[0005] The radio frequency devices may support communication over a broad range of frequencies, and a system-in-package of the radio frequency devices may include multiple chips to support wireless communications over the broad range of frequencies. By way of example, the system-in-package may include chips performing digital functions, as well as chips with analog circuits (e.g., little to no digital functions), on the same substrate. Moreover, the system-in-package may support a broad range of frequencies including frequencies used by multiple wireless carriers in various countries. Thus, to provide coverage over the broad range of frequencies or to add additional range coverage to the initial system-in-package, the system-in-package may include multiple chips and components. By way of example, the number of chips and components and/or the size of the chips may correlate to the range of frequencies supported for the wireless communications. Indeed, some of the chip sizes may be relatively large to support more frequencies. However, fitting these large chips in the system-in-package of the radio frequency devices may take up space and undesirably increase the size of the radio frequency devices.

SUMMARY

[0006] A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

[0007] In one embodiment, a radio frequency package may include a baseband processor, a transceiver, and a memory. The baseband processor may perform processing for wireless communication functions. Moreover, the transceiver may transmit and receive wireless signals based on the processing of the wireless communication functions. Additionally, the memory is associated with the baseband processor and may store instructions for performing the processing, in which the memory and the baseband processor are disposed on top of the transceiver.

[0008] In another embodiment, a stacked system package includes a first layer, a second layer, a third layer, and a fourth layer. The first layer may be composed of multiple layers of interconnects including high density vias, using coreless, cored, or redistribution layers. The high density vias may include vias (e.g., redistribution layer vias) having a size of approximately 5 or more micrometers (m) (e.g., between 5 to 10 m). The second layer is disposed on top of the first layer and the second layer may include one or more transceivers that transmit and receive wireless signals. Additionally or alternatively, the second layer may include one or more power management units that control power to or manage power for various chips. The second layer may also include through mold vias that route signals, provide shielding, or both. The through mold vias may be composed of mold and/or other materials, such as epoxy, molding dielectrics, laminate, and the like. By way of example, the through mold vias may be completely or partially composed of copper, epoxy, and molding material. The through mold vias may include copper studs, solder covered copper spheres, and the like. Additionally, the third layer is disposed on top of the second layer. The third layer may include a baseband processor that processes radio frequency functions and a memory that stores instructions for performing the process. Furthermore, the fourth layer is disposed on top of the third layer and may include a thermal gel that dissipates heat.

[0009] In some embodiments, a top side of the transceivers in the second layer may be disposed facing down towards a main logic board of the radio frequency device or facing up towards the third layer with the baseband processor. The top side may refer to a side including transceiver circuitry, interconnections that connect the circuitry, or both. In some embodiments, the transceivers may include through silicon vias for high density interconnects (e.g., interconnects including high density vias of 5 or more m) providing routing throughout the layers (e.g., the second layer) with the transceiver circuitry. In additional or alternative embodiments, the baseband processor may be disposed at the fourth layer or on top of the fourth layer in a fifth layer (e.g., top layer of the system package), for example, if the baseband processor is relatively too large to fit in the initial space of the third layer. As such, the large baseband processor may have more space for placement within the layer, as well as have at least one side that is exposed to components and materials external to the system package, such as mold. In such embodiments, the baseband processor may attach to components or materials that dissipate or transfer heat, such as a thermal heat sink, thermal gel, and the like. The top layer that includes electric components (e.g., the third layer) may also include one or more decoupling capacitors, bypass capacitors, or both, that may facilitate in maintaining power delivery above a threshold (e.g., maximum or approximately maximum power delivery provided by the radio frequency device). Furthermore, one or more integrated passive devices, magnetic components, or both, for radio frequency passive and digital passive functions may be disposed on top of, below, or between dies and/or packages of the radio frequency system package.

[0010] In yet another embodiment, a radio frequency package includes a baseband chip, a radio frequency chip, and a memory. The baseband chip performs processing for wireless communication functions. Moreover, the radio frequency chip transmits and receives wireless signals based on the processing of the wireless communication functions. Furthermore, the memory chip is associated with the baseband chip and stores instructions for performing the processing, in which the memory chip is disposed on top of the baseband chip.

[0011] Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well.

These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended only to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

[0013] FIG. 1 is a block diagram of an electronic device, according to an embodiment of the present disclosure;

[0014] FIG. 2 is a perspective view of a notebook computer representing an embodiment of the electronic device of FIG. 1;

[0015] FIG. 3 is a front view of a handheld device representing another embodiment of the electronic device of FIG. 1;

[0016] FIG. 4 is a front view of another handheld device representing another embodiment of the electronic device of FIG. 1;

[0017] FIG. 5 is a front view of a desktop computer representing another embodiment of the electronic device of FIG. 1;

[0018] FIG. 6 is a front view and side view of a wearable electronic device representing another embodiment of the electronic device of FIG. 1;

[0019] FIG. 7 is a schematic diagram of a radio frequency system package of the electronic device of FIG. 1, according to embodiments of the present disclosure;

[0020] FIG. 8 is a schematic diagram of the radio frequency system package of FIG. 7 with a shielding barrier, according to embodiments of the present disclosure;

[0021] FIG. 9 is a schematic diagram of a radio frequency system package with integrated passive devices, according to embodiments of the present disclosure;

[0022] FIG. 10 is a schematic diagram of a radio frequency system package with a continuous trench shielding, according to embodiments of the present disclosure;

[0023] FIG. 11 is a schematic diagram of a radio frequency system package with the trench shielding and through mold vias, according to embodiments of the present disclosure;

[0024] FIG. 12 is a schematic diagram of a radio frequency system package with the trench shielding and ferrite powder shielding, according to embodiments of the present disclosure;

[0025] FIG. 13 is a top view diagram of a bottom layer of a radio frequency system package with through mold vias, according to embodiments of the present disclosure;

[0026] FIG. 14 is another top view diagram of a bottom layer of a radio frequency system package with through mold vias, according to embodiments of the present disclosure;

[0027] FIG. 15 is a top view diagram of a top layer of a radio frequency system package, according to embodiments of the present disclosure;

[0028] FIG. 16 is a schematic diagram of a radio frequency system package with thermal paths, according to embodiments of the present disclosure;

[0029] FIG. 17 is a schematic diagram of signal communication paths within a radio frequency system package, according to embodiments of the present disclosure;

[0030] FIG. 18 is a schematic diagram of a radio frequency system package with an electromagnetic interference spacer, according to embodiments of the present disclosure;

[0031] FIG. 19 is a schematic diagram of a radio frequency system package with a partial through

mold via, according to embodiments of the present disclosure;

[0032] FIG. **20** is a schematic diagram of a radio frequency system package with a wire bonded memory, according to embodiments of the present disclosure;

[0033] FIG. **21** is a top view diagram of a top layer of a radio frequency system package with decoupling capacitors, according to embodiments of the present disclosure; and

[0034] FIG. **22** is another top view diagram of a bottom layer of a radio frequency system package with a ferrite enclosure, according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

[0035] When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, or “some embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Use of the term “approximately” or “near” should be understood to mean including close to a target (e.g., design, value, amount), such as within a margin of any suitable or contemplatable error (e.g., within 0.1% of a target, within 1% of a target, within 5% of a target, within 10% of a target, within 25% of a target, and so on).

[0036] As previously mentioned, radio frequency devices may include multiple packages including a radio frequency system package (e.g., system-in-package). The radio frequency system package may include multiple chips, such as a baseband processor, a memory associated with the baseband processor, and one or more transceivers (e.g., one or more radio frequency chips), as well as power management units (PMUs). The baseband processor may perform processing of radio frequency functions for communicating wireless signals, such as transmission signals and reception signals. The memory (e.g., dynamic random access memory) may store instructions, data, or algorithms for the baseband processor to perform the processing. The transceiver may be controlled by the baseband processor to transmit the transmission signals and receive the reception signals. For example, the memory and the baseband processor primarily perform digital functions or processing while the transceiver primarily performs analog functions or includes analog circuits to communicate the wireless signals.

[0037] The size of the transceiver chip may generally correspond to the range of frequencies supported for wireless communications. For example, that size of the transceiver may be relatively larger to support a broad range of frequencies than the size of a transceiver supporting a narrow range of frequencies. Moreover, the transceiver size may correspond to the number of cellular carriers supported, as well as the range of frequencies supported for multiple countries so that the same radio frequency devices may be used across many countries. For example, the size of the transceiver may be relatively larger to support multiple carriers and/or communication over frequencies in multiple countries than the size of a transceiver supporting few carriers and/or a few frequencies for a particular country (e.g., only the United States).

[0038] Often, the baseband processor, the memory associated with the baseband processor, and the transceiver may be placed on the same substrate within the radio frequency system package. However, the area of the substrate may not increase to support the large transceiver. Moreover, since the transceiver includes primarily analog circuits, reducing the size of the transceiver may be difficult while supporting the broad range of frequencies. On the other hand, the baseband processor may be reduced in size even with increasing functionality (e.g., may be relatively more scalable). Furthermore, analog circuits of the transceiver may be more susceptible to noise from other components within the radio frequency system package (e.g., digital or switching noise generated by the baseband processor).

[0039] As such, the present disclosure provides techniques for packaging that accommodates wireless communication over a broad range of frequencies while conserving space in the

packaging. In particular, the radio frequency system package includes a three-dimensional (3D) structure in which the baseband processor and the memory are stacked on top of the transceiver. In this manner, the transceiver may fit on the substrate, as well as add free space on the substrate for additional components and/or packages. Moreover, since the baseband processor performs primarily digital processing, the baseband processor may generate digital noise (e.g., switching noise). The digital noise may impact the transceiver (e.g., analog circuits of the transceiver) and cause signal degradation for the wireless communications. As such, the transceiver may be placed beneath the baseband processor in the stacked packaging (e.g., baseband processor at a top floor of the package and the receiver at a bottom floor). In some embodiments, memory may be placed on top of the baseband processor so that the stacked packaging includes the baseband processor on top of the transceiver, and the memory on top of the baseband processor.

[0040] Additionally, by placing the baseband processor on the top floor of the packaging, the baseband processor may consume more power and generate heat. The radio frequency system package described herein may include thermal gels, exposed dies (e.g., exposed top side of the baseband processor), and/or dummy dies to absorb or dissipate the heat. The radio frequency system may also include through mold vias that may include materials for providing shielding. The through mold vias may also route signals, as well as dissipate heat. Moreover, the radio frequency system package may include wire bonds that provide interconnections between components within the package, as well as provide shielding. Similarly, the radio frequency system package may include a shielding barrier of shielding layers (e.g., redistribution layers) that provide shielding, as well as route signals. In some embodiments, the radio frequency system package may include decoupling capacitors (e.g., a network of decoupling capacitors) to decouple noise from components (e.g., reduce or remove switching noise from the baseband processor). In this manner, the radio frequency system package described herein supports wireless communications over a broad range of frequencies while conserving space, manages thermal dissipation, ensures signal integrity, and provides noise immunity by shielding.

[0041] Turning first to FIG. 1, an electronic device **10** according to an embodiment of the present disclosure may include, among other things, one or more processor(s) **12**, memory **14**, nonvolatile storage **16**, a display **18**, input structures **22**, an input/output (I/O) interface **24**, a network interface **26**, a power source **28**, and a transceiver **30**. The various functional blocks shown in FIG. 1 may include hardware elements (including circuitry), software elements (including computer code stored on a computer-readable medium) or a combination of both hardware and software elements. It should be noted that FIG. 1 is merely one example of a particular implementation and is intended to illustrate the types of components that may be present in electronic device **10**.

[0042] By way of example, the electronic device **10** may represent a block diagram of the notebook computer depicted in FIG. 2, the handheld device depicted in FIG. 3, the handheld device depicted in FIG. 4, the desktop computer depicted in FIG. 5, the wearable electronic device depicted in FIG. 6, or similar devices. It should be noted that the processor(s) **12** and other related items in FIG. 1 may be generally referred to herein as “data processing circuitry.” Such data processing circuitry may be embodied wholly or in part as software, hardware, or any combination thereof. Furthermore, the processor(s) **12** and other related items in FIG. 1 may be a single contained processing module or may be incorporated wholly or partially within any of the other elements within the electronic device **10**.

[0043] In the electronic device **10** of FIG. 1, the processor(s) **12** may be operably coupled with a memory **14** and a nonvolatile storage **16** to perform various algorithms or instructions. For example, algorithms for dynamically selecting a range of frequencies for particular wireless communications may be saved in the memory **14** and/or nonvolatile storage **16**. The selection of the range of frequencies may be based on the bandwidth for particular applications executing on the electronic device **10**, the type of wireless communication (e.g., Bluetooth, global positioning system (GPS), Wi-Fi, and so forth), the country in which the electronic device **10** is being used

(e.g., United States, United Kingdom, Japan, etc.), and so forth. Such algorithms or instructions executed by the processor(s) **12** may be stored in any suitable article of manufacture that includes one or more tangible, computer-readable media. The tangible, computer-readable media may include the memory **14** and/or the nonvolatile storage **16**, individually or collectively, to store the algorithms or instructions. The memory **14** and the nonvolatile storage **16** may include any suitable articles of manufacture for storing data and executable instructions, such as random-access memory, read-only memory, rewritable flash memory, hard drives, and optical discs. In addition, programs (e.g., an operating system) encoded on such a computer program product may also include instructions that may be executed by the processor(s) **12** to enable the electronic device **10** to provide various functionalities.

[0044] In certain embodiments, the display **18** may be a liquid crystal display (LCD), which may display images generated on the electronic device **10**. In some embodiments, the display **18** may include a touch screen, which may facilitate user interaction with a user interface of the electronic device **10**. Furthermore, it should be appreciated that, in some embodiments, the display **18** may include one or more light-emitting diode (LED) displays, organic light-emitting diode (OLED) displays, active-matrix organic light-emitting diode (AMOLED) displays, or some combination of these and/or other display technologies.

[0045] The input structures **22** of the electronic device **10** may enable a user to interact with the electronic device **10** (e.g., pressing a button to increase or decrease a volume level). The I/O interface **24** may enable the electronic device **10** to interface with various other electronic devices, as may the network interface **26**. The network interface **26** may include, for example, one or more interfaces for a personal area network (PAN), such as a BLUETOOTH® network, for a local area network (LAN) or wireless local area network (WLAN), such as an 802.11x WI-FI® network, and/or for a wide area network (WAN), such as a 3.sup.rd generation (3G) cellular network, universal mobile telecommunication system (UMTS), 4.sup.th generation (4G) cellular network, long term evolution (LTE®) cellular network, long term evolution license assisted access (LTE-LAA) cellular network, 5.sup.th generation (5G) cellular network, and/or New Radio (NR) cellular network. In particular, the network interface **26** may include, for example, one or more interfaces for using a Release-15 cellular communication standard of the 5G specifications that include the millimeter wave (mmWave) frequency range (e.g., 30-300 GHz). The transceiver **30** of the electronic device **10**, which includes the transmitter and the receiver, may allow communication over the aforementioned networks (e.g., 5G, Wi-Fi, LTE-LAA, and so forth).

[0046] The network interface **26** may also include one or more interfaces for, for example, broadband fixed wireless access networks (e.g., WIMAX®), mobile broadband Wireless networks (mobile WIMAX®), asynchronous digital subscriber lines (e.g., ADSL, VDSL), digital video broadcasting-terrestrial (DVB-T®) network and its extension DVB Handheld (DVB-H®) network, ultra-wideband (UWB) network, alternating current (AC) power lines, and so forth.

[0047] In some embodiments, the electronic device **10** communicates over the aforementioned wireless networks (e.g., WI-FI®, WIMAX®, mobile WIMAX®, 4G, LTE®, 5G, and so forth) using the transceiver **30**. The transceiver **30** may include circuitry useful in both wirelessly receiving the reception signals at the receiver and wirelessly transmitting the transmission signals from the transmitter (e.g., data signals, wireless data signals, wireless carrier signals, radio frequency signals). Indeed, in some embodiments, the transceiver **30** may include the transmitter and the receiver combined into a single unit, or, in other embodiments, the transceiver **30** may include the transmitter separate from the receiver. The transceiver **30** may transmit and receive radio frequency signals to support voice and/or data communication in wireless applications such as, for example, PAN networks (e.g., BLUETOOTH®), WLAN networks (e.g., 802.11x WI-FI®), WAN networks (e.g., 3G, 4G, 5G, NR, and LTE® and LTE-LAA cellular networks), WIMAX® networks, mobile WIMAX® networks, ADSL and VDSL networks, DVB-T® and DVB-H® networks, UWB networks, and so forth. As further illustrated, the electronic device **10** may include

the power source **28**. The power source **28** may include any suitable source of power, such as a rechargeable lithium polymer (Li-poly) battery and/or an alternating current (AC) power converter. [0048] In certain embodiments, the electronic device **10** may take the form of a computer, a portable electronic device, a wearable electronic device, or other type of electronic device. Such computers may be generally portable (such as laptop, notebook, and tablet computers), or generally used in one place (such as desktop computers, workstations, and/or servers). In certain embodiments, the electronic device **10** in the form of a computer may be a model of a MacBook®, MacBook® Pro, MacBook Air®, iMac®, Mac® mini, or Mac Pro® available from Apple Inc. of Cupertino, California. By way of example, the electronic device **10**, taking the form of a notebook computer **10A**, is illustrated in FIG. **2** in accordance with one embodiment of the present disclosure. The depicted notebook computer **10A** may include a housing or enclosure **31**, a display **18**, input structures **22**, and ports of an I/O interface **24**. In one embodiment, the input structures **22** (such as a keyboard and/or touchpad) may be used to interact with the computer **10A**, such as to start, control, or operate a graphical user interface (GUI) and/or applications running on computer **10A**. For example, a keyboard and/or touchpad may allow a user to navigate a user interface and/or an application interface displayed on display **18**.

[0049] FIG. **3** depicts a front view of a handheld device **10B**, which represents one embodiment of the electronic device **10**. The handheld device **10B** may represent, for example, a portable phone, a media player, a personal data organizer, a handheld game platform, or any combination of such devices. By way of example, the handheld device **10B** may be a model of an iPhone® available from Apple Inc. of Cupertino, California. The handheld device **10B** may include an enclosure **31** to protect interior components from physical damage and/or to shield them from electromagnetic interference. The enclosure **31** may surround the display **18**, which displays an array of icons **19**. By way of example, when an icon **19** is selected either by an input structure **22** or a touch sensing component of the electronic display **18**, an application program may launch. The I/O interfaces **24** may open through the enclosure **31** and may include, for example, an I/O port for a hardwired connection for charging and/or content manipulation using a standard connector and protocol, such as the Lightning connector provided by Apple Inc. of Cupertino, California, a universal serial bus (USB), or other similar connector and protocol. The I/O interfaces **24** may be associated with wiring and connectors within the radio frequency packaging of the electronic device **10**.

[0050] The input structures **22**, in combination with the display **18**, may allow a user to control the handheld device **10B**. For example, the input structures **22** may activate or deactivate the handheld device **10B**, navigate user interface to a home screen, a user-configurable application screen, and/or activate a voice-recognition feature of the handheld device **10B**. Other input structures **22** may provide volume control, or may toggle between vibrate and ring modes. The input structures **22** may also include a microphone that may obtain a user's voice for various voice-related features, and a speaker that may enable audio playback and/or certain phone capabilities. The input structures **22** may also include a headphone input that may provide a connection to external speakers and/or headphones.

[0051] FIG. **4** depicts a front view of another handheld device **10C**, which represents another embodiment of the electronic device **10**. The handheld device **10C** may represent, for example, a tablet computer, or one of various portable computing devices. By way of example, the handheld device **10C** may be a tablet-sized embodiment of the electronic device **10**, which may be, for example, a model of an iPad® available from Apple Inc. of Cupertino, California.

[0052] Turning to FIG. **5**, a computer **10D** may represent another embodiment of the electronic device **10** of FIG. **1**. The computer **10D** may be any computer, such as a desktop computer, a server, or a notebook computer, but may also be a standalone media player or video gaming machine. By way of example, the computer **10D** may be an iMac®, a MacBook®, or other similar device by Apple Inc. of Cupertino, California. It should be noted that the computer **10D** may also represent a personal computer (PC) by another manufacturer. A similar enclosure **31** may be

provided to protect and enclose internal components of the computer **10D**, such as the display **18**. In certain embodiments, a user of the computer **10D** may interact with the computer **10D** using various peripheral input structures **22**, such as the keyboard **22A** or mouse **22B** (e.g., input structures **22**), which may connect to the computer **10D**.

[0053] Similarly, FIG. **6** depicts a wearable electronic device **10E** representing another embodiment of the electronic device **10** of FIG. **1** that may be configured to operate using the techniques described herein. By way of example, the wearable electronic device **10E**, which may include a wristband **23**, may be an Apple Watch® by Apple Inc. of Cupertino, California. However, in other embodiments, the wearable electronic device **10E** may include any wearable electronic device such as, for example, a wearable exercise monitoring device (e.g., pedometer, accelerometer, heart rate monitor), or other device by another manufacturer. The display **18** of the wearable electronic device **10E** may include a touch screen display **18** (e.g., LCD, LED display, OLED display, active-matrix organic light emitting diode (AMOLED) display, and so forth), as well as input structures **22**, which may allow users to interact with a user interface of the wearable electronic device **10E**.

[0054] With the foregoing in mind, FIG. **7** is schematic diagram of a radio frequency system package **50** of the electronic device **10** of FIG. **1**. Although the depicted embodiment shows a three-dimensional (3D) stack of chips, which represents a particular embodiment, the radio frequency system package **50** may include one or more of the chips positioned in a different stack structure (e.g., two-dimensional (2D) or 2.5D). For example, one or more of the chips may be positioned on top, beneath, and/or adjacent to another one of the one or more chips, for example, along a z-axis **29** (e.g., height), an x-axis **27** (e.g., width), and/or a y-axis **25** (e.g., length) in a 3D space.

[0055] In the depicted embodiment, the radio frequency system package **50** includes multiple chips, such as a first transceiver **30A** (Transceiver **1**), a second transceiver **30B** (Transceiver **2**), a memory **52**, and a baseband processor **54** (e.g., cellular modem). The chips may couple to (e.g., directly or indirectly connected) to a printed circuit board (PCB) substrate (not shown) (e.g., a main logic board). Although the following descriptions describe the radio frequency system package **50** with two transceivers **30**, which represents a particular embodiment, the radio frequency system package **50** described herein may additionally or alternatively include one or more transceivers **30**. The transceivers **30** may couple to an antenna array that includes multiple antennas that transmit and/or receive wireless signals. The antenna array may form a directional beam using signals emitted by each of the antennas (e.g., beamforming). The first transceiver **30A** and the second transceiver **30B** may each provide the transceiver **30** functions described herein or the functions may be distributed between the first transceiver **30A** and the second transceiver **30B**. Moreover, a top side (e.g., a side that is disposed proximate the antennas of the radio frequency device, a non-silicon bulk side, a portion including circuits and/or interconnects, or any combination thereof) of the first transceiver **30A** and the second transceiver **30B** may be disposed as facing down towards a printed circuit board on which the radio frequency system package **50** is disposed, or facing up towards the baseband processor **54** to facilitate a high interconnect density between layers of the radio frequency system package **50**. The package of the transceivers **30** may include dies with or without the through silicon vias, as previously discussed. The through silicon vias may provide increased interconnection density (e.g., a size of approximately 5 or more m).

[0056] In general, the transceivers **30** may primarily include analog circuits and perform analog functions, such as performing frequency modulation to transmit and receive wireless signals. The analog functions may involve utilizing capacitors, inductors, and the like for analog functions. In some embodiments, the transceivers **30** may also perform some digital functions, such as but not limited to sending and receiving binary data. On the other hand, the baseband processor **54** may primarily perform digital functions related to processing data for the wireless communications. In some embodiments, the baseband processor **54** may couple to or be integrated with an application processor. The baseband processor **54** may communicate with the memory **52** and the transceivers **30** to carry out the wireless communications. For example, the transceivers **30** may receive data

from one or more panels (e.g., printed circuit boards that include one or more packages, such as system-in-packages that include substrates, dies, modules, and/or components) that may correspond to a particular side or portion of the radio frequency device. The transceiver **30** may send the received data to the baseband processor **54** for processing.

[0057] The baseband processor **54** may include one or more microprocessors, one or more “general-purpose” microprocessors, one or more special-purpose microprocessors, and/or one or more application specific integrated circuits (ASICs), or some combination thereof. For example, the baseband processor **54** may include one or more reduced instruction set (RISC) processors. In some instances, the baseband processor **54** may perform processing (e.g., execute software programs and/or instructions) for specification functions, such as specific functions related to specific wireless communication. The specific functions may include receiving or generating wireless signals, selecting particular antennas for transmitting or receiving signals of the antenna array, selecting an amplification level to amplify transmission signals using a power management circuitry, determining gain of the wireless signals transmitted and/or received from a particular transmitter and/or receiver associated with a particular antenna of the antenna array, and so forth. In some instances, the baseband processor **54** may be integrated with the processor **12** and perform additional functions related to the wireless communications, such as functions related to the display **18**, adjusting bandwidth consumption, and so forth.

[0058] The baseband processor **54** may communicate with the memory **52** for processing instructions to perform the functions related to wireless communications. The memory **52** may store information such as control software, configuration data, and so forth. The baseband processor **54** may include particular baseband firmware, such as a baseband operating system. In some embodiments, the baseband processor **54** may perform digital functions or mostly digital functions, such as converting digital data into radio frequency signals (and vice-versa) that can subsequently be transmitted from the transceivers **30** over a network.

[0059] The memory **52** may include a tangible, non-transitory, machine-readable-medium, such as a volatile memory (e.g., a random access memory (RAM)) and/or a nonvolatile memory (e.g., a read-only memory (ROM)), flash memory, a hard drive, or any other suitable optical, magnetic, or solid-state storage medium, or a combination thereof. In some embodiments, the memory **52** may be static random access memory (SRAM) or dynamic random access memory (DRAM). The memory may store a variety of information and may be used for various purposes. For example, the memory device may store machine-readable and/or processor-executable instructions (e.g., in the form of software or a computer program) for the baseband processors **54** to execute, such as instructions for processing received data and transmitting or receiving signals at the transceivers **30**.

[0060] A transceiver **30**, as previously discussed with respect to FIG. **1**, is a device that includes a transmitter and a receiver in a single package, and may transmit and receive data via wireless signals communicated on particular radio frequency using the antennas of the antenna array. Specifically, the transceivers **30** may each include a transmitter and a receiver that include components that facilitate transmission and reception of wireless signals, such as those sent and received between electronic devices **10** using mmWave communication technology or any other suitable communication protocol. By way of example, when communicating on the mmWave frequencies, an electronic device **10** may utilize beamforming techniques to form the directional beam, as previously mentioned. The transmitters of the transceivers **30** may include one or more phase shifters, transmitter power detectors, and power amplifiers. The transmitter phase shifters may modulate (e.g., phase-shift) transmission signals (e.g., wireless signals transmitted from antennas of the antenna array) and may form a beam that may be steered in a particular direction (e.g., the directional beam), such as towards another electronic device (e.g., an electronic device **10**, a base station). The power amplifiers may amplify power level of transmission signals. Specifically, the power amplifiers may be supplied with a power amplifier supply voltage to control

the amount of amplification provided by the power amplifiers (e.g., increase or decrease amplification, which may affect the antenna gain at the corresponding antennas). The transmitter power detectors may measure power of the transmission signals sent from the antennas of the antenna array. In some embodiments, the transceivers **30** may receive digital data signals from the baseband processor **54** and convert the digital data signals from a baseband protocol to analog signals, which may be modulated to phase-shift keying signals (e.g., quadriphase-shift keying (QPSK)), pulse-amplitude modulation (PAM) signals, quadrature amplitude modulation (QAM) signals, binary phase shift keying (BPSK) signals, to 5G signals, 6G signals, and the like. In some instances, the transceiver **30** may upconvert signals and downconvert signals, such as down converting mmWave signals to intermediate frequencies (IF) signals.

[0061] The receivers of the transceivers **30** may include one or more receiver phase shifters, low noise amplifiers, and receiver power detectors. The receiver phase shifters and the receiver power detectors may function similarly to the transmitter phase shifters and the transmitter power detectors. The low noise amplifiers may amplify the power level of reception signals (e.g., wireless signals received at antennas of the antenna array). Additional components in the transmitter and/or the receiver may include, but are not limited to, filters, mixers, and/or attenuators.

[0062] As previously discussed, integrating multiple chips that may increase in size within the same particular area within a package may be difficult. Specifically, one or more chips may increase in size to fit additional circuitry or components to provide certain functionalities. However, the area allotted for the chips and/or associated chips may not increase (e.g., initial area for the radio frequency system package **50** before adding additional components and circuitry). For example, a designer may increase the size of the transceiver **30** chip to add additional functionalities, such as for compatibility to communicate over a broader range of frequencies. However, it may be desired that the area within the radio frequency system package **50** to place the transceiver **30** and associated chips not increase (e.g., maintain original dimensions or decreased dimensions). As such, the radio frequency system package **50** includes the memory **52** and the baseband processor **54** on top of the transceivers **30** rather than the memory **52** and/or the baseband processor **54** being placed adjacent to the transceivers **30** on the same substrate (e.g., same layer of printed circuit board). As will be described herein, the radio frequency system package **50** may efficiently accommodate the co-located chips (e.g., the memory **52**, the baseband processor **54**, and the transceivers **30**) by stacking the chips while conserving space (e.g., along the x-axis **27** and/or the y-axis **25**). By stacking the chips in a 3D structure, the area initially including the memory **52** and the baseband processor **54** in the 2D structure may become available for larger transceivers **30** and/or other components for additional functionality. Another advantage of stacking chips is that the chip-to-chip communication may use relatively less power than other chip communication methods or standards, such as M-PHY, PCIe, USB, and the like. Thus, the stacked chip connections may also provide more power efficient communication. Although the following descriptions describe space conservation packaging techniques applied to a particular system package **50** related to the transceivers **30**, the baseband processor **54**, and the memory **52** associated with the baseband processor **54**, the techniques may also apply to the other chips and/or packages (e.g., packages including multiple chips) of the electronic device **10**.

[0063] FIG. **8** is a schematic diagram of the radio frequency system package **50** of FIG. **7** with a shielding barrier. Although FIGS. **8-22** depict particular embodiments of the systems and methods described herein, and the following discussions describe the particular embodiments with respect to the corresponding figures, the features described for any one of the particular embodiment may additionally or alternatively apply to another embodiment. That is, a feature described in FIG. **8** may apply to the embodiment depicted and described for FIG. **9**, FIG. **11**, FIG. **21**, and so forth. The radio frequency system package **50** may be described as having one or more layers in the 3D stacked structure, such as a first layer **62A**, a second layer **62B**, a third layer **62C**, and a fourth layer **62D** (e.g., each having one or more layers). The first layer **62A** may include a first shielding layer

60A (Shielding Layer 1). The second layer **62B** that is positioned on top of the first layer **62A** may include one or more transceivers **30**, such as the first transceiver **30A**, the second transceiver **30B**, one or more integrated passive devices **70** (IPDs), one or more through mold vias (TMVs) **64**, and a second shielding layer **60B** (Shielding Layer 2). Additionally or alternatively, the second layer **62B** may include one or more power management units (not shown) that control power to or manage power for various chips. The third layer **62C** that is positioned on top of the second layer **62B**, may include a high-density substrate **61** (e.g., redistribution layer, coreless substrate of two or fewer layers, and the like), the memory **52**, the baseband processor **54**, a thermal gel **66**, and one or more decoupling capacitors **68**. In some embodiments, the high-density substrate **61** may include routing for power and/or logic. High-density substrate **61** may optionally include one or more structures embedded in the substrate material to facilitate power and/or logic routing, such as a communication bar or interface bar. In some embodiments, the baseband processor **54** and the memory **52** may be arranged within the third layer **62C** laterally adjacent to one another. Moreover, and as shown, decoupling capacitors **58** may be disposed laterally adjacent to the baseband processor **54**, the memory **52**, or both. In additional or alternative embodiments, passive devices (e.g., integrated passive devices **70**) may be disposed laterally adjacent to the baseband processor **54**, the memory **52**, or both, such as to provide decoupling, insulation from noise, and the like. In some embodiments, the memory **52** and the baseband processor **54** are encapsulated by mold **76** (e.g., mold compound) or another material to provide insulation, rigidity or other structural enhancements to the third layer **62C**. In some embodiments, the mold **76** or other encapsulating material may be planarized and/or selectively removed after application to expose one or more sides of components in the third layer **62C**, such as the top of the baseband processor **54**. The fourth layer **62D** that is positioned on top of the third layer **62C** may include the thermal gel **66**. In some embodiments, as shown in FIG. 8, the thermal gel **66** may be placed on an exposed surface of a component in the third layer **62C** (e.g., top side of baseband processor **54**). In some embodiments, some thermal gel **66** may be included in the third layer **62C** (e.g., as shown on top of memory **52**), and may be at least partially encapsulated in mold **76** or another encapsulating material.

[0064] The first shielding layer **60A** and the second shielding layer **60B** may form a shielding barrier between the transceivers **30** and the third layer **62C** that includes the memory **52** and the baseband processor **54**. As will be discussed herein, the first shielding layer **60A** and the second shielding layer **60B** may also provide high density routing (e.g., through vias). By way of example, the first layer **62A** may be composed of multiple layers of interconnects including high density vias, such as through mold vias, using coreless, cored, or redistribution layers. The through mold vias may be composed of mold and/or other materials, such as metal. By way of example, the through mold vias may be completely or partially composed of copper. The through mold vias may include copper studs, solder covered copper spheres, and the like. In some embodiments, through mold vias may have a first diameter at one end of the length of a through mold via and a second diameter at the opposite end of the length of the through mold via. The first diameter of the through mold via may differ in size from the second diameter. In some embodiments, to minimize insertion loss, one or more through mold vias may be positioned to provide a path of relatively minimal distance between points of contact (e.g., pads, solder balls, pins). Generally, the chips, components, and/or layers with the chips and/or components of the radio frequency system package **50** may be coupled directly and/or indirectly by solder balls or pads **63**, as shown. For example, the solder balls **63** may be disposed between components of the system package **50**. The solder balls **63** may include solder (e.g., lead, tin, epoxy, low-melting alloy, and/or the like) that provides electrical connections between components to facilitate a communication path between the components. For example, the second layer **62B** may couple to the third layer **62C** via the solder balls **63**.

[0065] The shielding layers **60** may include layers of metallic material that absorbs radio and electromagnetic waves. For example, the metallic material may include copper and the like. In some embodiments, the shielding layers may include one or more layers, such that they are

redistribution layers (RDLs). The redistribution layers may include one or more metal layers (e.g., a copper metal layer) on top of one or more chips, such as the transceivers **30**, to form and provide input/output (I/O) pads available for access to the chips via the pads. For example, the metal layers may form interconnects that electrically connect one part of the radio frequency system package **50** to another area of the radio frequency system package **50** or other packages of the electronic device **10** (e.g., mmWave package). As such, the shielding layers **60** may provide routing and/or shielding for the chips connected to the shielding layers **60**. By way of example, the shielding layers **60** may provide routing and/or shielding for the transceiver **30**, a high-density substrate **61**, the memory **52** coupled to the high-density substrate **61**, the baseband processor **54** coupled to the high-density substrate, and so forth). The high-density substrate **61** may include a substrate of one or more layers (e.g., one, three, four, and so forth) of insulating material (e.g., copper) that provides electrical insulation. The high-density substrate **61** may include an embedded trace substrate (ETS), an Ajinomoto build-up film (ABF), a redistribution layer (RDL), and the like. In some embodiments, the high-density substrate **61** may include one or more integrated passive devices **70** in the substrate for decoupling noise. That is, additionally or alternatively to the thermal gel **66**, the integrated passive device **70** may be stacked to provide the decoupling. Moreover, in some instances, a thin film or trench capacitors may be placed under the memory **52** along the z-axis **29** to provide decoupling for the memory **52**.

[0066] In some embodiments, the memory **52** may be grounded for additional shielding, such as by wire bonds **72** connecting the thermal gel **66** and/or the memory **52** to the high-density substrate **61**. The wire bonds **72** may include one or more metal wires, (e.g., each wire diameter of approximately 10 to 15 micrometers (m)) such as aluminum, copper, silver, gold, and the like metals, for providing interconnections between components. For example, the wire bonds **72** may make interconnections between a chip (e.g., integrated circuit) or other semiconductor device and its packaging, connect an integrated circuit to another electronic device, connect a printed circuit board to another, and so forth.

[0067] Moreover, the through mold vias **64** may provide shielding and routing for the chips of the radio frequency system package **50**. Generally, vias are electrical connections between layers of a package so that signals may be routed between the layers of the package. For example, the vias may include holes that are drilled through two or more stacked layers and the holes are plated with metal (e.g., copper) that form the electrical connections through the insulation that separates the layers. Through mold vias **64** may create the interconnecting vias through mold caps. The through mold vias **64** may include insulating material, such as the copper, which may provide shielding. As such, the shielding provided from the shielding layers **60** and the through mold vias **64** surrounding the transceivers **30** may form a shielding enclosure around the transceivers **30** (e.g., a fence-like shield enclosure). The transceivers **30** may be generally sensitive to noise, and thus, this shielding enclosure may shield the transceivers **30** from external noise from other chips and components of the radio frequency system package **50**. For example, the enclosure may shield the transceivers **30** from digital noise from the baseband processor **54** that may otherwise negatively impact radio frequency communications from the transceiver **30** (e.g., decrease signal-to-noise ratio (SNR)). As such, the through mold vias **64** may provide both signal routing and shielding. In some embodiments, and as previously mentioned, a top side of the transceivers **30** in the second layer **62B** (e.g., a side that includes the antennas) may be disposed facing down towards a main logic board of the radio frequency device or facing up towards the third layer **62C** with the baseband processor **54**. In some embodiments, the transceivers **30** may include through silicon vias (TSVs) for high density interconnects providing routing throughout the layers (e.g., the second layer **62B**) with the transceiver circuitry.

[0068] In some instances, by placing the memory **52** and/or the baseband processor **54** on top of the transceivers **30** in the 3D radio frequency system package rather than adjacently, the memory **52** and/or the baseband processor **54** may consume more power than when placed adjacently to the

transceivers **30** on the same layer (e.g., in 2D). Thus, the third layer **62C** and the fourth layer **62D** include the thermal gel **66** on top of the memory **52** and the baseband processor **54**, respectively. The thermal gel **66** may facilitate heat dissipation such that heat from the memory **52** and the baseband processor **54** caused by the increased power consumption may dissipate through the thermal gel **66**. Although the radio frequency system package **50** described herein uses the thermal gel **66** for heat dissipation, which is a particular embodiment, the radio frequency system package **50** may additionally or alternatively use exposed dies (e.g., exposing the top of the memory **52** (e.g., die) rather than placing the thermal gel **66** on top of the memory **52**). Additionally or alternatively to the thermal gel **66**, the radio frequency system package **50** may include dummy dies placed on top of or beneath (e.g., below) the memory **52** and/or the baseband processor **54**. In some embodiments, as previously mentioned, the baseband processor **54** may be disposed on the fourth layer **62D** or on top of the fourth layer **62D** in a fifth layer (not shown) (e.g., top layer of the system package **50**), for example, if the baseband processor **54** is relatively too large to fit in the initial space of the third layer **62C**. As such, the large baseband processor **54** may have more space for placement within the layer, as well as have at least one side that is exposed to components and materials external to the radio frequency system package **50**, such as mold. In such embodiments, the baseband processor **54** may attach to components or materials that dissipate or transfer heat, such as the thermal gel **66**, a thermal heat sink, and the like.

[0069] Additionally or alternatively to the shielding barrier of the shielding layers **60**, the decoupling capacitors **68** of the third layer **62C** may include one or more capacitors that isolate or decouple one part of an electrical network from another, for example, to reduce noise. For example, noise caused by components may be shunted through the capacitor, reducing the effect it has on the radio frequency system package **50**. Generally, the decoupling capacitors **68** may be placed as close as possible to the source of the signal being decoupled (e.g., within a threshold distance). Here, to decouple the memory **52** and the baseband processor **54** from the transceivers **30**, the decoupling capacitors **68** may be placed next to, approximately next to, or adjacent to the memory **52** and the baseband processor **54**. In this manner, the decoupling capacitors **68** may reduce or prevent noise caused by the memory **52** and/or the baseband processor **54** from negatively impacting the transceivers **30** (e.g., decreasing signal-to-noise ratio and decreasing signal integrity). Additionally or alternatively to the decoupling capacitors **68**, the third layer **62C** may include bypass capacitors. The decoupling capacitors **68**, the bypass capacitors, or both, may facilitate in maintaining power delivery above a threshold (e.g., maximum or approximately maximum power delivery provided by the radio frequency device. In some embodiments, ferrite beads and integrated passive devices **70** (e.g., inductors) may be used to isolate noise (e.g., noise occurring over high frequencies) from the baseband processor **54** to a radio frequency integrated circuit of the radio frequency device.

[0070] Generally, integrated passive devices, magnetic components, or both, for radio frequency passive and digital passive functions, may be disposed on top of, below, or between dies and/or packages of the radio frequency system package **50**. To illustrate, the radio frequency system package **50** in the depicted embodiment includes one or more integrated passive devices **70**. Integrated passive devices **70** (e.g., integrated passive components and/or embedded passive components) include electronic components, such as one or more of resistors, capacitors, inductors, microstrip lines, impedance matching elements, baluns, filters, and the like. The one or more components are integrated into the same package or substrate (e.g., resistor-capacitor-inductor (RCL) network) to form a resistive silicon substrate. In some embodiments, the integrated passive devices **70** may be packed into a die or chip, or be stacked (e.g., on top of another die or chip) in 3D with active integrated circuits or other integrated passive devices **70**. The integrated passive devices **70** may reduce power loss and enhance signal transmission. By coupling the transceivers **30** to the integrated passive devices **70** (e.g., here, the integrated passive device **70** is connected to the second transceiver **30B**), signal loss from the transceivers **30** may be reduced.

[0071] FIG. **9** is a schematic diagram of a radio frequency system package **50** with integrated

passive devices **70** under the first shielding layer **60A**. As shown, in some embodiments, the radio frequency system package **50** may also include integrated passive devices **70** positioned under the first shielding layer **60A** to reduce power loss and enhance signal transmission from the transceivers **30**. These integrated passive device **70** include components and function as previously described. By way of example, the integrated passive devices **70** may include capacitors that reduce digital noise from the baseband processor **54**, such as digital noise or switching noise (e.g., noise from components used for performing digital functions by the baseband processor **54**). For example, the switching noise caused by enabling wireless communications over varying frequencies (e.g., converting signals), and the noise may occur at a particular frequency. This noise may otherwise reduce radio frequency signal quality (e.g., decrease signal-to-noise ratio) from the transceivers **30**.

[0072] The radio frequency system package **50** may also include additional solder balls **63**. These solder balls **63** may couple the first shielding layer **60A** to a printed circuit board (e.g., another die or chip, a main logic board, and the like). The integrated passive devices **70**, such as the integrated passive devices **70** adjacent to the solder balls **63** under the first shielding layer **60A**, may provide shielding from noise caused by components on the printed circuit board or coupled to the printed circuit board.

[0073] FIG. **10** is a schematic diagram of a radio frequency system package **50** with continuous shielding around chips. The radio frequency system package **50** also includes multiple decoupling capacitors **68** to form a decoupling network for filtering noise. In particular, a first decoupling capacitor **68A**, a second decoupling capacitor **68B**, a third decoupling capacitor **68C**, and a fourth decoupling capacitor **68D** form a decoupling capacitor network around the baseband processor **54** to decouple noise generated from the baseband processor **54**. The decoupling capacitors **68** may be connected via one or more wire bonds **72**. For example and as shown, the first decoupling capacitor **68A** may connect to the third decoupling capacitor **68C** via a first wire bond **72A**. The second decoupling capacitor **68B** may connect to the fourth decoupling capacitor **68D** via a second wire bond **72B**. The wire bonds **72A** and **72B** may include one or more wires. The decoupling capacitors **68** connected by the wire bonds **72** around the baseband processor **54** may provide a local filtering (e.g., local to the transceiver **30** and/or the radio frequency system package **50**). Although the depicted embodiment shows the decoupling capacitor network of the decoupling capacitors **68** around the baseband processor **54**, which represents a particular embodiment, the radio frequency system package **50** may include multiple (e.g., two, four, six, and so forth) decoupling capacitors **68** around any components of the radio frequency system package **50** that may generate noise and adversely impact integrity of another component of the radio frequency system package **50** (e.g., signal integrity of signals from the transceivers **30**).

[0074] To form the continuous shielding around chips, the radio frequency system package **50** includes trenches **74** around the chips. In some embodiments, the trenches **74** may be formed around mold **76** that encapsulates the chips while in additional or alternative embodiments, the trenches **74** may be formed directly around the chips. By way of example, although FIG. **10** depicts a trench **74** formed around a mold **76** that encapsulates transceivers **30**, which represents a particular embodiment, the discussions described herein may also apply to a trench **74** formed directly around the transceivers **30**. The trenches **74** may be filled with conformal shielding material, such as sputtered metal, metallic materials, paste materials, metal foam, and the like. The conformal shielding material may include copper, brass, nickel, silver, steel, tin, and the like metals. Additionally or alternatively, the trench may be filled with metal nano-ferrite composite fillers to provide isolation from noise (e.g., occurring at low MHz to high GHz frequencies). The trenches **74** are formed around the chips in a continuous manner or semi-continuous manner, such as without a break or approximately without a break, so that the shielding material fills the trenches **74** in a manner to provide continuous or approximately continuous electromagnetic shielding around the chips (e.g., continuous shielding or semi-continuous shielding). For example, the

trenches **74** may be formed on each exposed side of a chip. If the sides of the chips are not exposed, the trench **74** may be formed at a point where the next closest side of the radio frequency system package **50** is exposed.

[0075] By way of example, a trench **74** may be formed on top of the baseband processor **54** and the memory **52**, along the x-axis **27** and along the sides of the fourth layer **62D** of the radio frequency system package **50** along the negative z-axis **29** (e.g., when the baseband processor **54** and the memory **52** are viewed as being at the center in the 3D space (0, 0, 0)). For example, the trench **74** may not be formed along the sides of the baseband processor **54** along the negative z-axis **29** since the decoupling capacitors **68** are placed there, and thus, the sides are not exposed. On the other hand, the first transceiver **30A** and the second transceiver **30B** are placed next to each other, such that they may be connected (e.g., indirectly or directly coupled) and the outer sides of the transceivers **30** along the negative z-axis **29** are exposed. As such, a trench **74** may be formed around the exposed top side of the transceivers **30** along the x-axis **27** and the exposed outer sides of the transceivers **30** along the negative z-axis **29**. The trenches **74** may not be formed at the bottom sides of the baseband processor **54**, the memory **52**, and the transceivers **30** since these chips are stacked in the radio frequency system package **50** and the bottom side is not exposed. In some embodiments, the trenches **74** may be formed on an exposed side of the chip based on a threshold space available between the side and a closest substrate or component to the side. Noise immunity and reduced interference may be achieved by forming the continuous shielding around the primarily digital processing chips, such as the baseband processor **54** and the associated memory **52**, and around the primarily analog chips such as the transceivers **30**. For example, noise from the shielded primarily digital processing chips and noise from the shielded primarily analog chips may not impact each other (e.g., reduced impact of noise).

[0076] The radio frequency system package **50** may also include the through mold vias **64** surrounded by mold **76**. The through mold vias **64** and the network of decoupling capacitors **68** may facilitate providing an intended power voltage to the baseband processor **54**. In particular, the through mold vias **64** may facilitate routing the power and reducing any current-resistor (IR) voltage drop before it reaches baseband processor **54**. The through mold vias **64** may include shielded material (e.g., metal). Moreover, to ensure that the power to the baseband processor **54** is decoupled, the network of decoupling capacitors may filter the voltage before the voltage goes into the baseband processor **54**. Additionally, the conformal shielding may reduce noise produced from the through mold vias **64** and/or that may otherwise travel to and impact the transceivers **30**. In some embodiments, the transceivers **30** may be shielded by trenches **74**, through mold vias **64**, or both. For example, a semi-continuous shielding may include conformal shielding in a trench **74** along at least a first exposed side of the first transceiver **30A** and at least one through mold via **64** along at least a second exposed side of the first transceiver **30A**.

[0077] In general, as voltage is provided in the radio frequency system package **50**, such as vertically or laterally, components providing the voltage may generate noise. In some instances, the components may generate noise in various directions, functioning as an antenna radiating signals in the various direction. By way of example, when providing high power voltage signals that may switch in a switch-mode power supply, the switching voltage signals may result in switching power noise. As such, the decoupling capacitors **68** may filter out the noise before the baseband processor **54** receives the voltage. In some embodiments, the trenches **74** or a portion of the trenches **74** may be substituted by the through mold vias **64**, as shown in FIG. **11**. For example, rather than creating the continuous shielding around the chips using the trenches **74** filled with conformal shielding material, the shielding may include portions that include trenches **74** with the shielding material and/or the through mold vias **64** made of similar shielding materials and/or providing a similar shielding effect (e.g., similar shielding capabilities as the shielding materials in the trenches **74**).

[0078] Additionally or alternatively to the trenches **74** with conformal shielding material and/or the through mold vias **64** for providing shielding, the radio frequency system package **50** may include

ferrite powder **78** or the like, as shown in FIG. **12**. For example, additionally or alternatively to the ferrite powder **78**, the radio frequency system package **50** may include an electromagnetic interference (EMI) absorber (e.g., EMI absorber film). In the depicted embodiment, a layer of ferrite powder **78** may be placed above a trench **74** of the conformal shielding. In some embodiments, a gap or space between components and chips may be included. In the current embodiment, the transceivers **30** may be positioned at a distance from the high-density substrate **61**, forming a gap or a space between the transceivers **30** and the high-density substrate **61**. Rather than filling the entire gap with mold, trenches **74** with conformal shielding and/or a layer of ferrite powder **78** may be used to fill at least a portion of the gap and provide shielding. In some embodiments, the trenches **74** may vary in depth (e.g., deeper for more conformal shielding material when there is a large gap). In some embodiments, using trenches **74** and/or the ferrite powder **78** may be based on a threshold dimension of the gap. By way of example, in the depicted embodiment, the gap may be approximately 50 to 70 m deep along the z-axis **29** while the conformal shielding of the trench **74** is approximately 5 to 10 m deep. As such, a layer of ferrite powder **78** may be added to the conformal shielding to fill in the gap and provide shielding. Although the depicted embodiment shows the ferrite powder **78** placed on top of the trench **74**, which represents a particular embodiment, the ferrite powder **78** may be placed elsewhere around the components of the radio frequency system package **50**, such as to also form continuous shielding (e.g., similar placement as the trenches **74** with the conformal shielding materials). The ferrite powder **78** may include EMI shielding materials, such as iron oxide blended with one or more metallic elements. The metallic elements may include strontium, barium, manganese, nickel, zinc, and the like.

[0079] The combination of the trench **74** filled with conformal shielding and the layer of ferrite powder **78** may provide shielding from noise over a broad range of frequencies. In particular, the conformal shielding material may shield from interference occurring over high frequencies while the ferrite powder **78** may shield from interference occurring at low frequencies. For example, the ferrite powder **78** shields from noise traveling at low frequencies while being permeable (e.g., enabling to pass through) at high frequencies. Thus, noise may be attenuated or substantially may not travel from the third layers **62C** with the baseband processor **54** to the second layer **62B** with the transceivers **30**, which may be relatively more sensitive to noise than the baseband processor **54** or other components of the radio frequency system package **50**. As such, the trenches **74** filled with conformal shielding and layers of ferrite powder **78** may provide signal integrity and noise immunity for the chips of the radio frequency system package **50**.

[0080] FIG. **13** is a top view diagram of a bottom layer of the radio frequency system package **50** of the electronic device of FIG. **8** with through mold vias **64**. The bottom layer (e.g., the second layer **62B** of FIG. **12**) may include the first transceiver **30A** and the second transceiver **30B**. As shown in FIG. **13**, the transceivers **30** may be of the same or approximately similar dimensions, such that the horizontal side along the x-axis **27**, the vertical side along the y-axis **25**, or both, of the transceivers **30** line up. On the other hand, and as shown in FIG. **14**, the dimensions may be different, such that the horizontal side, the vertical side, or both, of the transceiver **30** do not line up when placed adjacently. In both FIG. **13** and FIG. **14**, the transceiver **30** are surrounded by through mold vias **64**. Although the following descriptions describe a particular number of rows and a particular number of columns of the through mold vias **64**, which represents particular embodiments, the radio frequency system package **50** may include one or more rows, one or more columns, or both, with one or more through mold vias **64** (e.g., more rows and columns correspond to more shielding). Additionally, although the following descriptions describe the through mold vias **64** including a particular number of the through mold vias **64** as including shielding material or providing signal routing, which represents particular embodiments, the radio frequency system package **50** may include any number of the through mold vias **64** as including shielding material, providing signal routing, or both.

[0081] For example, the through mold vias **64** may generally provide signal routing between layers **62** of the radio frequency system package **50**, provide shielding, or both. Furthermore, although the through mold vias **64** are positioned along the sides of the transceivers **30** along the y-axis **25**, which represents a particular embodiment in which the top and bottom sides along the x-axis **27** are not exposed (e.g., due to other substrates, chips, and so forth in the stacked radio frequency system package **50**), the transceivers **30** may be fully surrounded by or enclosed by the through mold vias **64** if the particular sides are exposed.

[0082] As shown in both FIG. **13** and FIG. **14**, the transceivers **30** may be surrounded by through mold vias **64** (indicated by the unfilled circles) that provide shielding and are grounded. The through mold vias **64** that provide shielding may be surrounded by through mold vias **64** that route signals (indicated by filled circles). The spatial arrangement of through mold vias **64** providing shielding and grounding and the through mold vias **64** providing routing may be based on the function provided by the respective through mold vias **64**. By way of example, the through mold vias **64** providing shielding may connect to analog ground and not digital ground and may be arranged accordingly.

[0083] The arrangement of the through mold vias **64** for shielding and the through mold vias **64** for routing signals may be based on a predetermined level of intended shielding and/or for particular frequencies, destination of the signals, signals paths, and other system package related features. Although the following descriptions describe the through mold vias **64** used for either shielding or routing signals, which represents a particular embodiment, the radio frequency system package **50** may include through mold vias **64** that provide shielding as well as route signals.

[0084] As shown, the through mold vias **64** around the transceivers **30** are of high-density (e.g., many through mold vias **64** per area). For example, the through mold vias **64** may be placed in close proximity to each other, effectively forming a shielding enclosure or a faraday cage for the transceivers **30** (e.g., within a threshold distance to provide shielding function of the faraday cage). For example, the density may provide equal to or less than a 0.25 millimeter (mm) pitch (e.g., center to center distance between two through molds vias **64** is less than 0.25 mm). The outer through mold vias **64** (e.g., outside the shielding through mold vias **64**) may route signals, such as from the transceivers **30** to the baseband processor **54** (e.g., from the bottom floor to a top floor of the radio frequency system package **50**, along the positive z-axis **29** of FIG. **12**). Additionally, the frequency shielding provided by the depicted faraday cages of FIG. **13** and FIG. **14** may provide shielding in a frequency range of up to approximately 40-80 gigahertz (GHz) (e.g., 40 GHz).

[0085] FIG. **15** is a top view diagram of a top layer (e.g., third layer **62C** of FIG. **12**) of a radio frequency system package **50**. The memory **52** may include one or more banks or rows of event data record (EDR) memory or the like. In general, the memory **52** may use approximately 5-10 gigabits (GB) (e.g., 8 GB) of memory to process the incoming signals. The radio frequency system package **50** may include one or more decoupling capacitors **68** (e.g., varying in size and/or capacitor values) that are placed around the exposed side or areas of the baseband processor **54** and the memory **52**. The decoupling capacitors **68** may function as previously described (e.g., provide a decoupling network).

[0086] By way of example, the area including the memory **52** and the baseband processor **54** may have a height of approximately 5 to 10 millimeters (mm) along the y-axis **25** and a width of approximately 10 to 15 mm along the x-axis **27** (e.g., 7 mm by 13 mm, 5 mm by 10 mm, and so forth). The baseband processor **54** may have a height and width of approximately 5 to 7 mm (e.g., 6 mm×6 mm, 6.1 mm×6.4 mm, and so forth). The memory **52** chip may have a height and width of approximately 3 to 5 mm (e.g., 4.4 mm×3.1 mm, 5.14 mm×4.6 mm, 5.2 mm×5.0 mm, and so forth). By way of another example, the memory **52** and the baseband processor **54** may have depths along the z-axis **29** of approximately 0.5 to 1.0 mm (e.g., 0.7 mm, 0.8 mm, 0.9 mm, and so forth).

[0087] The baseband processor **54** and the memory **52** may be connected or coupled using wire bonds **72** (as shown), chip-to-chip connections, or both. In some embodiments, the baseband

processor **54** and the memory **52** may be indirectly electrically and/or communicatively coupled using an intermediate routing structure such as an interposer, communication bar or logic bar, with direct connections between the routing structure and the baseband processor **54** and direct connections between the routing structure and the memory **52**. The wire bonds **72** and/or the chip-to-chip connections may send data between the memory **52** and the baseband processor **54** (e.g., send data from the memory **52** to the baseband processor **54**). In some embodiments, the wire bonds **72** may be replaced to remove any possible latency issues, high bandwidth issues, and so forth, otherwise occurring with the wire bonds **72**. In the chip-to-chip embodiments, the chip-to-chip connection between the baseband processor **54** and the memory **52** may involve high speed serial links for communicating.

[0088] FIG. **16** is a schematic diagram of a radio frequency system package **50** with thermal paths. The radio frequency system package **50** may consume more power by stacking the memory **52** and the baseband processor **54** on top of the transceivers **30** within the radio frequency system package **50**, in a smaller x-axis **27** and y-axis **25** area. The large power consumption may generate heat and the baseband processor **54** may consume most of the generated heat. In one embodiment, the top side of the baseband processor **54** along the positive z-axis **29** may be exposed to dissipate the heat from the baseband processor **54** and the radio frequency system package **50**. In additional or alternative embodiments, and as shown, the top side of the baseband processor **54** may include the thermal gel **66**. Moreover, in additional or alternative embodiments, the top side of the baseband processor **54** may include a heat spreader (e.g., a graphite heat spreader), a dummy chip, or both to dissipate the heat. As shown by the solid line arrows, the heat may dissipate from the baseband processor **54** using these techniques, such as towards and through the thermal gel **66** (e.g., the heat removing technique). The exposed top, the thermal gel **66**, the dummy chip, and/or the heat spreader, may form a continuous heat spreader to remove the heat from the baseband processor **54**. [0089] In some embodiments, and as indicated by the dashed line arrows, heat generated from the baseband processor **54** may also travel in various directions, such as down into the bottom floors along a negative z-axis **29**. The heat may impact the transceiver **30**, as well as a main logic board **71** connected to the radio frequency system package **50**. For example, cross heating may occur in the radio frequency system package **50**. The through mold vias **64** may absorb or dissipate the heat. In particular, the through mold vias **64** may be made of heat dissipating materials, such as copper. As such, the through mold vias **64** discussed herein may provide signal filtering, reduce noise by providing shielding, and also transfer heat out of the radio frequency system package **50** (e.g., function as a heat conduit for transferring heat). In particular, the amount of metal for the through mold vias **64** may correspond to the amount of shielding and heat removal (e.g., greater amount of metal corresponds to more shielding and greater amount of heat removed).

[0090] FIG. **17** is a schematic diagram of signal communication and signal paths **80** within a radio frequency system package **50**. As indicated by the arrows, the signal paths **80** may include a signal flow from the baseband processor **54** down along the negative z-axis **29** to the transceivers **30**, and from the transceivers **30** to the printed circuit board (e.g., main logic board **71**, other boards, packages, or components on the printed circuit board). In particular, the signal path **80** from the baseband processor **54** may include transmission related signals and reception related signals communicated using a serial communication protocol. The baseband processor **54** may also communicate peripheral Component Interconnect Express (PCIe) signals, signals related to memory (DDR), and signals related power (PWR) to the printed circuit board (e.g., boards, packages, or components on the printed circuit board). In some embodiments, power signals may be communicated through the through mold vias **64** that may function as power buses. As shown, the transceivers **30** may send high speed input/output (I/O) signals, low speed input/output signals (I/O), general-purpose input/output (GPIO) signals, debug signals (Debug), clock signals (CLK), and the like. In some embodiments, the transceivers **30** may communicate a different number of signals, different types of signals, and/or at different speeds. By way of example, the first

transceiver **30A** may communicate **30** to **50** signals simultaneously (e.g., indicated by the two arrows from the first transceiver **30A**) while the second transceiver **30B** communicates **10** to **20** signals simultaneously (e.g., indicated by the single arrow from the second transceiver **30B**). As such, the signals may be communicated in a top-down manner within the radio frequency system package **50**. In additional or alternative embodiments, at least some of the signals may be communicated laterally, in a bottom-up manner along the signal paths **80** (e.g., from the printed circuit board to transceivers **30**, from the transceivers **30** to the baseband processor **54**, and so forth), or both.

[0091] FIG. **18** is a schematic diagram of a radio frequency system package **50** with an electromagnetic interference (EMI) spacer **90**. In some embodiments, to conserve space along the x-axis **27**, the y-axis **25**, or both, the memory **52** may be stacked on top of the baseband processor **54** rather than placed adjacently. By way of example, the radio frequency system package **50** may be included in the wearable electronic device **10E** of FIG. **6**. Generally, the wearable electronic device **10E** may not communicate over a broad range of frequencies (e.g., limited to GPS, Wi-Fi, a few cellular frequencies, and so forth) and as such, the transceiver **30** may be smaller than a radio frequency system package **50** of other devices that communicate over a broader range of frequencies, such as the handheld device **10B** of FIG. **3**. Moreover, the dimensions of the wearable electronic device **10E** may be relatively small and thus, may utilize the radio frequency system package **50** conserving space along the x-axis **27** and the y-axis **25** by stacking the memory **52** on top of the baseband processor **54**. However, this stacked structure may be used for any suitable radio frequency devices (e.g., devices **10A-10E**).

[0092] In some embodiments, and as shown, the baseband processor **54** may be integrated with or couple to a radio frequency chip **92** (e.g., chip with baseband processing and radio frequency functions combined). The radio frequency chip **92** may perform similar functions to the transceivers **30** described herein. The memory **52** may communicate with the baseband processor **54** using serial communication protocol (e.g., sending data one bit at a time, sequentially, over a communication channel), as well as use differential links for communicating with the baseband processor **54** integrated with the radio frequency chip **92**. In some instances, noise from the baseband processor **54** may negatively impact signal integrity from the memory **52** and noise from the memory **52** may negatively impact signal integrity from the baseband processor **54**. In particular, since the memory **52** is stacked on top of the baseband processor **54**, the memory **52** may be relatively larger along the x-axis **27**, the y-axis **25**, or both (e.g., approximately the same as the baseband processor **54**), rather than along the z-axis **29** to fit inside a small package (e.g., not too tall for a wearable device **10E**). In such embodiments, the semiconducting substrate of the memory **52** may generate noise that leaks into the radio frequency chip **92** of the baseband processor **54**, causing signal degradation of radio frequency signals. To reduce the impact of the leaked noise, the radio frequency system package **50** includes the electromagnetic interference spacer **90** between the memory **52** and the baseband processor **54**. The electromagnetic interference spacer **90** may include a substrate of shielding material that provides shielding across a range of frequencies used for communication by the radio frequency system package **50**. Furthermore, the radio frequency system package **50** may include wire bonds **72** to ground the radio frequency system package **50**. As shown, the wire bonds **72** may interconnect the memory **52**, the electromagnetic interference spacer **90**, and the shielding layer **60**, effectively providing a shielding enclosure (e.g., a faraday cage).

[0093] FIG. **19** is a schematic diagram of a radio frequency system package **50** with a partial through mold via **65**. The partial through mold via **65** may provide similar functions as the through mold vias **64**. To form the partial through mold via **65**, metal may be sputtered into the partial through mold via **65** and connected to shielding of one or more of the chips. For example, the partial through mold via **65** may connect to wire bonds **72**, which connect to a shielding layer **60** (e.g., the first shielding layer **60A** and the second shielding layer **60B**). As such, the radio

frequency system package **50** described herein may be shielded on each side of the package using trenches **74** with conformal shielding, ferrite powder **78**, partial through mold vias **65**, wire bonds **72** connecting to ground, and so forth.

[0094] In some embodiments, the wire bonds **72** may connect to the top of the package (e.g., top side of the memory **52**), which may further connect to other components (e.g., the partial through mold via **65**). The process for using the wire bonds **72** for shielding may involve grinding down polished metal wires so that the raw metal is exposed and connecting the exposed metal wires to a shield (e.g., shielding substrate). For example, and as shown, the wire bonds **72** may connect the memory **52** to the shielding layer **60** to provide shielding for the memory **52**. Thus, the wire bonds **72** may provide interconnections, shielding for the radio frequency system package **50**, or both (e.g., interconnecting shielding). In this manner, the partial through mold via **65** and the wire bonds **72** may shield the radio frequency chip **92** from noise generated from the baseband processor **54**, the memory **52**, or both.

[0095] FIG. **20** is a schematic diagram of a radio frequency system package **50** with a wire bonded memory **52**. In some embodiments and as shown, the memory **52** may be stacked on top of the baseband processor **54** rather than adjacently, for example, to conserve area along the x-axis **27** and the y-axis **25**. The radio frequency system package **50** may include a power management circuitry **96** (e.g., an integrated circuit on a chip or a power management package). The power management circuitry **96** may control functions related to power for transmitting the transmission signals and receiving the reception signals. The power management circuitry **96** couples (e.g., communicatively or indirectly via interconnects) to the radio frequency chip **92**.

[0096] As shown, the memory **52** may connect to the high-density substrate **61** via wire bonds **72**. The high-density substrate **61** may include laminate-based, organic-based, or both, shielding materials. This wire bonding may reduce or prevent noise from the memory **52**, the baseband processor **54**, or both, from traveling to and impacting components on the bottom floor of the radio frequency system package, such as the radio frequency chip **92**, the power management circuitry **96**, or both. The shielding layer **60** and the through mold vias **64** may provide additional shielding within the radio frequency system package **50**. The shielding layer **60** may include one or more metal layers (e.g., copper metal layer), as previously discussed.

[0097] FIG. **21** is a top view diagram of a top layer of a radio frequency system package **50** with decoupling components, such as decoupling capacitors **68**. As shown, the memory **52** is placed on top of the baseband processor **54** of the first radio frequency system package **50A** and the second radio frequency system package **50B**. Additionally, wire bonds **72** attached to the memory **52** may connect to other parts of the radio frequency system packages **50A**, **50B**, such as on lower layers of the radio frequency system packages **50A**, **50B** (not shown in the top view). The radio frequency system packages **50A**, **50B** may include decoupling capacitors **68** around the memory **52**, the baseband processor **54**, or both. The decoupling capacitors **68** may decouple noise and provide filtering, as previously discussed. The decoupling capacitors **68** of the first radio frequency system package **50A** may be relatively smaller than the decoupling capacitors **68** of the second radio frequency system package **50B**. The size of the decoupling capacitors **68** may be based on the dimensions of the top floor, the memory **52**, the baseband processor **54**, or both. In some embodiments, the top floor of the first radio frequency system package **50A** may include a length along the y-axis **25** and a width along the x-axis **27** of approximately 3 mm to 7 mm. By way of example, the top floor of the first radio frequency system package **50A** may be 6 mm×4 mm while the top floor of the second radio frequency system package **50B** may be 6.3 mm×4.3 mm.

[0098] FIG. **22** is a top view diagram of a bottom layer of a radio frequency system package **50** with a ferrite enclosure **98**. As shown, the bottom layer of the first and the second radio frequency system packages **50A**, **50B** may include the radio frequency chip **92**. The radio frequency chip **92** may be enclosed with through mold vias **64** around the radio frequency chip **92**. The through mold vias **64** may function as described with respect to FIG. **13** and FIG. **14**. The radio frequency system

packages **50** may also include decoupling capacitors **68** around the through mold vias **64**. The decoupling capacitors **68** may decouple noise and provide filtering, as previously discussed. The size of the decoupling capacitors **68** may be based on the dimensions of the bottom layer, the radio frequency chip **92**, or both. In some embodiments, the bottom layer of the first radio frequency system package **50A** may include a length along the y-axis **25** and a width along the x-axis **27** of approximately 3 mm to 7 mm. By way of example, the bottom layer of the first radio frequency system package **50A** may be 6 mm×4 mm while the top floor of the second radio frequency system package **50B** may be 6.3 mm×4.3 mm. As such, the techniques described herein facilitate wireless communication over a broad range of frequencies while conserving space in a radio frequency package, managing thermal dissipation, ensuring signal integrity, and providing noise immunity by shielding.

[0099] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ,” it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

[0100] It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

Claims

1. A stacked system package, comprising: a first layer comprising a transceiver configured to transmit and receive wireless signals; a second layer disposed on top of the first layer, the second layer comprising: a baseband processor configured to process radio frequency functions; a memory configured to store instructions for the baseband processor; and a high-density substrate configured to route signals associated with the radio frequency functions.
2. The stacked system package of claim 1, wherein the high-density substrate comprises a communication bar.
3. The stacked system package of claim 1, wherein the high-density substrate comprises one or more layers of insulating material.
4. The stacked system package of claim 1, wherein the high-density substrate comprises an embedded trace substrate (ETS), an Ajinomoto build-up film (ABF), a redistribution layer (RDL), or a combination thereof.
5. The stacked system package of claim 1, wherein the high-density substrate comprises one or more passive devices.
6. The stacked system package of claim 5, wherein the one or more passive devices are configured to decouple noise from the baseband processor, the memory, or both.
7. The stacked system package of claim 1, wherein the second layer comprises one or more wire bonds connecting the memory to the high-density substrate, the one or more wire bonds configured to ground the memory, route the signals associated with the radio frequency functions, or both.
8. The stacked system package of claim 1, wherein the first layer comprises one or more passive devices.
9. The stacked system package of claim 8, wherein the first layer comprises a shielding layer disposed between the transceiver and the one or more passive devices.

- 10.** The stacked system package of claim 1, wherein the second layer comprises one or more decoupling capacitors around the baseband processor configured to decouple noise generated from the baseband processor.
- 11.** The stacked system package of claim 10, wherein the high-density substrate comprises one or more wire bonds, and wherein the one or more decoupling capacitors are connected via the one or more wire bonds.
- 12.** A radio frequency package, comprising: a baseband processor; a transceiver configured to transmit and receive wireless signals based on the baseband processor; one or more integrated passive devices, wherein the transceiver is disposed on top of the one or more integrated passive devices in the radio frequency package; and a memory associated with the baseband processor, the memory configured to store instructions for the baseband processor, wherein the memory and the baseband processor are disposed on top of the transceiver in the radio frequency package.
- 13.** The radio frequency package of claim 12, wherein the one or more integrated passive devices are configured to reduce power loss associated with the transceiver transmitting and receiving the wireless signals.
- 14.** The radio frequency package of claim 12, wherein the one or more integrated passive devices are configured to reduce digital noise generated by the baseband processor.
- 15.** The radio frequency package of claim 12, comprising one or more solder balls configured to couple the radio frequency package to a printed circuit board, wherein the transceiver is disposed on top of the one or more solder balls.
- 16.** The radio frequency package of claim 15, wherein the one or more integrated passive devices are configured to shield the transceiver, the baseband processor, the memory, or a combination thereof from a noise signal generated by components on the printed circuit board.
- 17.** A radio frequency package, comprising: a first layer comprising a transceiver configured to transmit and receive wireless signals; a second layer disposed on top of the first layer, the second layer comprising a high-density substrate; and a third layer disposed on top of the second layer, the third layer comprising: a baseband chip configured to process radio frequency functions; and a memory configured to store instructions for the processing.
- 18.** The radio frequency package of claim 17, wherein the high-density substrate comprises a communication bar configured to route logic signals associated with the processing.
- 19.** The radio frequency package of claim 17, wherein the baseband chip and the memory are laterally adjacent within the third layer.
- 20.** The radio frequency package of claim 17, comprising one or more passive devices configured to shield the transceiver from digital noise from the baseband chip, noise signals from outside of the radio frequency package, or both, wherein the first layer is disposed on top of the one or more passive devices.
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